

Referee Report
Economic Modelling

Title: Fiscal Consolidations in Commodity-Exporting Countries: A Small Open Economy DSGE Perspective

ECMODE-D-25-02910

The authors use a DSGE model estimated on data for Ecuador to then assess the macroeconomic effects of the country's fiscal consolidation plan assisted by the IMF. The paper considers the importance of alternative paths for oil price, a key source of government revenue, in affecting the outcomes of the fiscal programme.

I enjoyed reading the paper and found it both interesting and well-presented. The derivation of the model, including its steady state and equilibrium equations blocks were very transparent and easy to follow. I hope that the authors will find the below comments useful in sharpening the contribution of the paper to the literature.

Comments:

1. Ecuador defaulted on its debt in 2008 and 2020, yet the model does not include any features that capture sovereign default risk (nor does the estimation take it into account). I believe that this is an important omission, especially given the aims of the paper. One rather tractable way of including default risk is that adopted by Corsetti et al. (2013) and Cantore et al. (2019); this approach would only involve a minor extension of the model but it would greatly enhance the paper.
2. One of the key contributions of the paper is to consider how alternative paths for oil prices could affect the fiscal consolidation programme adopted by Ecuador. As shown in Figure 4, the effects of higher or lower oil prices than in the benchmark are symmetric. This is not surprising given that the model is solved at first order and that there are no non-linearities present, such as borrowing constraints. One potential solution would be to solve the model under perfect foresight as this would permit preserving the original non-linearities in the model.
3. Following the previous comment, if I understood the authors correctly, they obtained the state space representation of the model and then fed the sequence of 'shocks' that are the new fiscal measures adopted by the consolidation programme. If that is the case, I do not think that this is the right approach as it does not take into account that the programme agreed with the IMF is known by the agents. Put differently, given an agreed path for the fiscal policy variables (and assuming that this is credible), agents' expectations will respond as soon as the new policy is known. Again, this can be easily implemented by solving the model under perfect

foresight for a given path of the exogenous variable so that the amount of work required is minimal (especially if the authors use Dynare for this). Nonetheless, the approach suggested above as well as the one used in the paper, assumes that the fiscal plan is fully credible. Given Ecuador's economic history, the paper would benefit from a brief discussion on how the possibility of a policy reversal could affect the outcomes in the model. My understanding is that the IMF Extended Fund Facility arrangement has been successful thus far but the comment pertains to the credibility of plan when announced, not to its actual delivery.

4. A key aspect of the analysis is that the fiscal consolidation is based on an Extended Fund Facility provided by the IMF, which involves financial assistance. Yes, external transfers are not present in the model and this is a key aspect of the agreement.
5. The model assumes that the transfer policy is endogenous and that it responds to the business cycle as shown in equation (13). Moreover, given the estimated value of α_{TR} , transfers are procyclical. While this may make sense during the estimation period, it is not clear that the government would maintain such policy during a fiscal consolidation programme.
6. On p. 39 the authors state that prior distributions for the parameters of the fiscal expenditure rules, such as ρ_x , are centred using values from external estimation using the same sample period as for the full model. However, Prior distributions either reflect subjective opinions or summarize information derived from data sets not included in the estimation sample. For example, one could use pre-sample data to inform the prior.
7. In presenting the impulse responses (IRFs) from the model, it is not clear whether the model is solved using the posterior mean (median?) estimates. Moreover, it might be desirable to plot the mean of the IRFs generated when parameters are drawn from the posterior distribution, as opposed to plotting the IRFs using the mean estimates, in order to convey a measure of uncertainty

Minor comments:

- In the abstract, the authors state that the model is calibrated and estimated using Ecuador's 2020–2025 IMF-supported programme but this is incorrect. The estimation considers the period 2004-2019) and this is used to assess the IMF programme 2020-25.
- Given that the one group of households cannot save and live hand-to-mouth, yet they optimally choose their labour supply, I think that labelling households as Ricardian and non-Ricardian is more appropriate. Denoting the former as 'optimiser' is slightly misleading as the latter also optimise, albeit only intratemporally.
- There is an odd break on p. 23.

- 'Definition' is often misspelled as 'Defenition' in the paper.
- The paper mentions the sample periods but the quarters are not specified.
- On p. 37 the authors state that some of the variables have been de-trended using the Hodrick-Prescott filter. This is inappropriate as it is a two-sided filter while the solution to the DSGE model takes the form of a state-space system (see Pfeifer). This is easily resolved by using the one-sided HP filter.

References

- Cantore, Cristiano, Paul Levine, Giovanni Melina, and Joseph Pearlman**, "Optimal Fiscal And Monetary Policy, Debt Crisis, And Management," *Macroeconomic Dynamics*, April 2019, *23* (3), 1166–1204.
- Corsetti, Giancarlo, Keith Kuester, André Meier, and Gernot J. Müller**, "Sovereign Risk, Fiscal Policy, and Macroeconomic Stability," *Economic Journal*, February 2013, *0*, 99–132.